1. As you walk around Beebe Lake, what sources of randomness do you notice? Use all five senses!

2. What sources of randomness are present but not directly accessible to your senses?

3. Treasure hunt! For each of the six distributions below:

   - Describe a random variable in the wild that has (approximately) that distribution.
   - Estimate the parameters (p or λ or μ, σ^2 or a, b).
   - State your assumptions.

   Tip: Scope your random variables precisely! Examples:
   Too vague: "X = the amount of water flowing over the dam"
   Better: "X = **liters of water per second flowing over the dam at noon on November 1**"
   Too vague: "Y = number of eggs in a bird's nest"
   Better: "Y = **current number of eggs in a uniformly random robin's nest in Ithaca**"

   (i) Bernoulli: Ber(p)

   (ii) Geometric: Geom(p)

   (iii) Exponential: Exp(λ)

(*continued on back*)

(iv) Poisson: Pois(λ)

(v) Normal: $N(\mu, \sigma^2)$

(vi) Uniform: Unif(a, b)

4. Wild card: Find a random variable in the wild that's <u>not</u> well modeled by any of the named distributions in the treasure hunt above. Is your random variable discrete or continuous? Describe its p.m.f. or p.d.f. or c.d.f.

5. Find two random variables in the wild and describe their <u>joint distribution</u>. Are they discrete or continuous? Are they positively or negatively correlated?